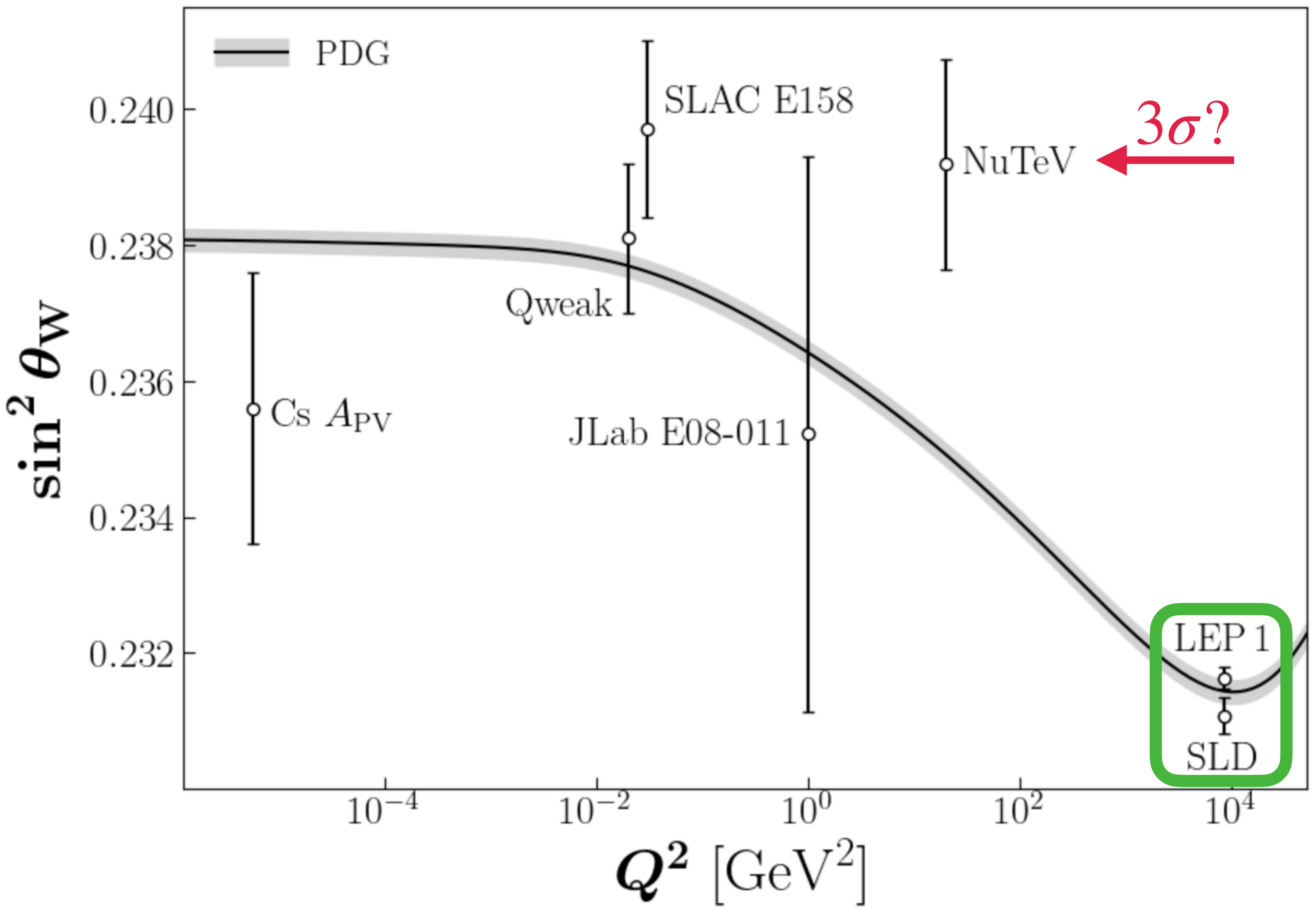
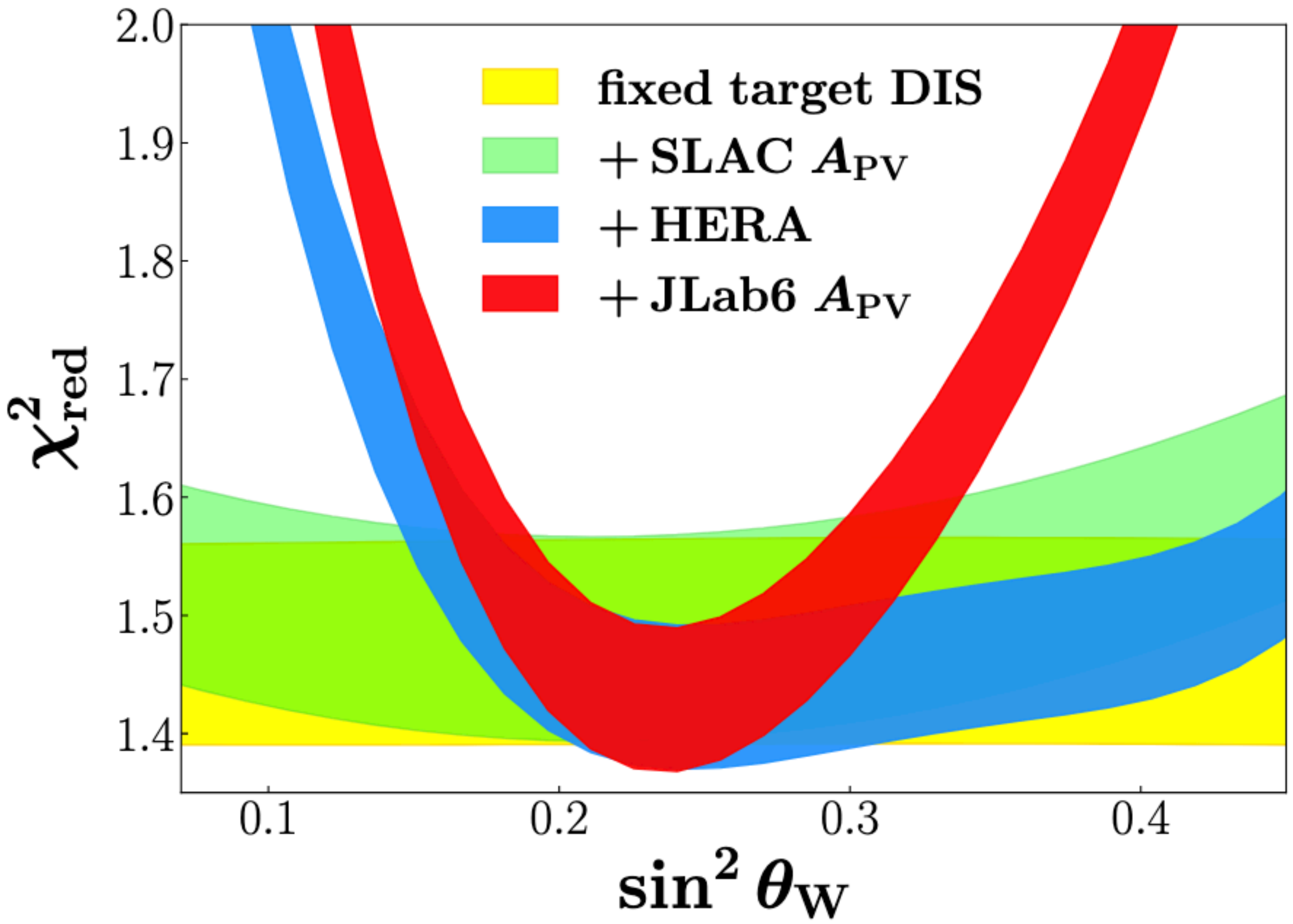
Impact of PVDIS on the weak mixing angle and high-x parton distributions

Richard Whitehill

Collaborators: W. Melnitchouk & N. Sato

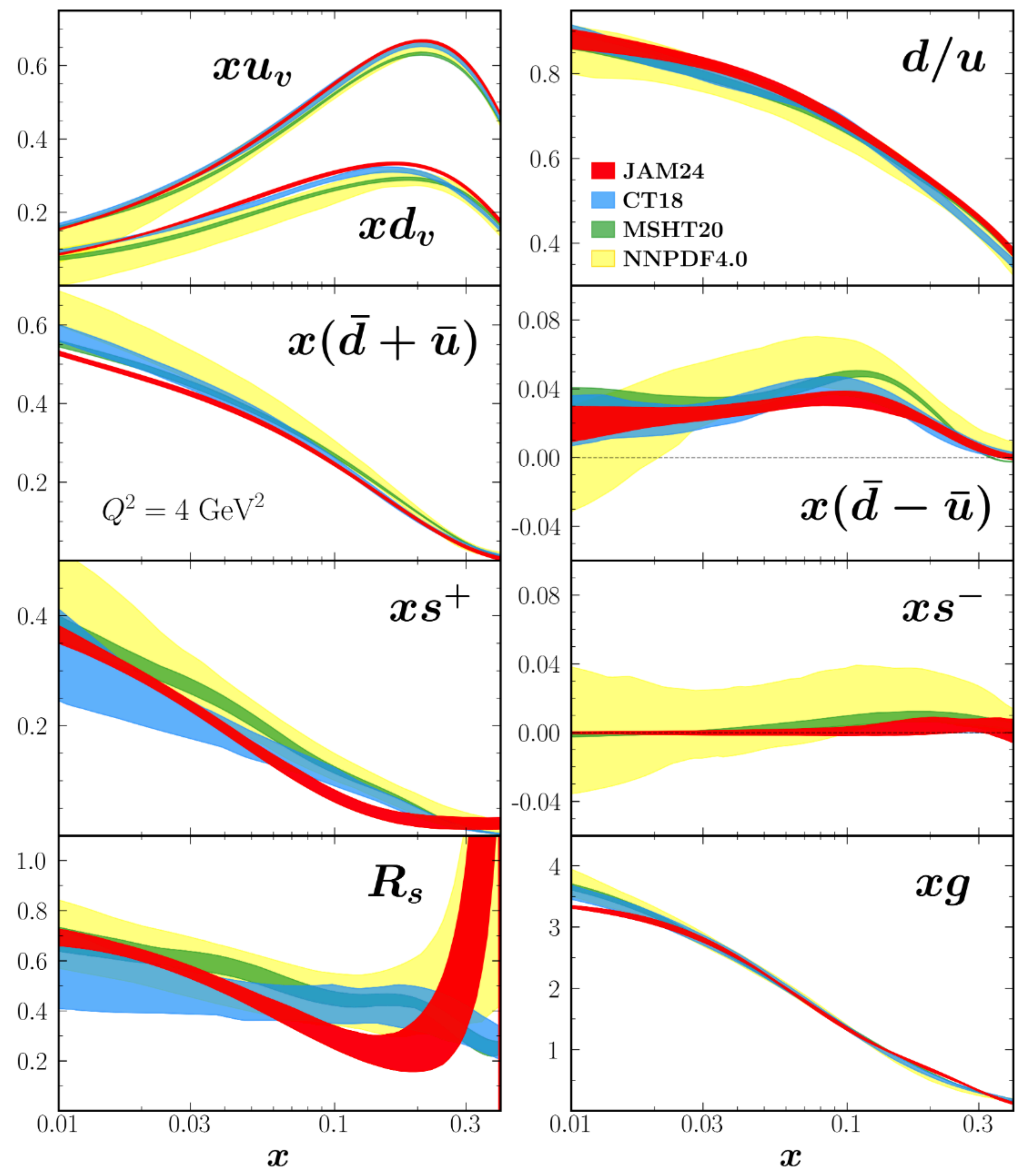
EW physics in global analysis



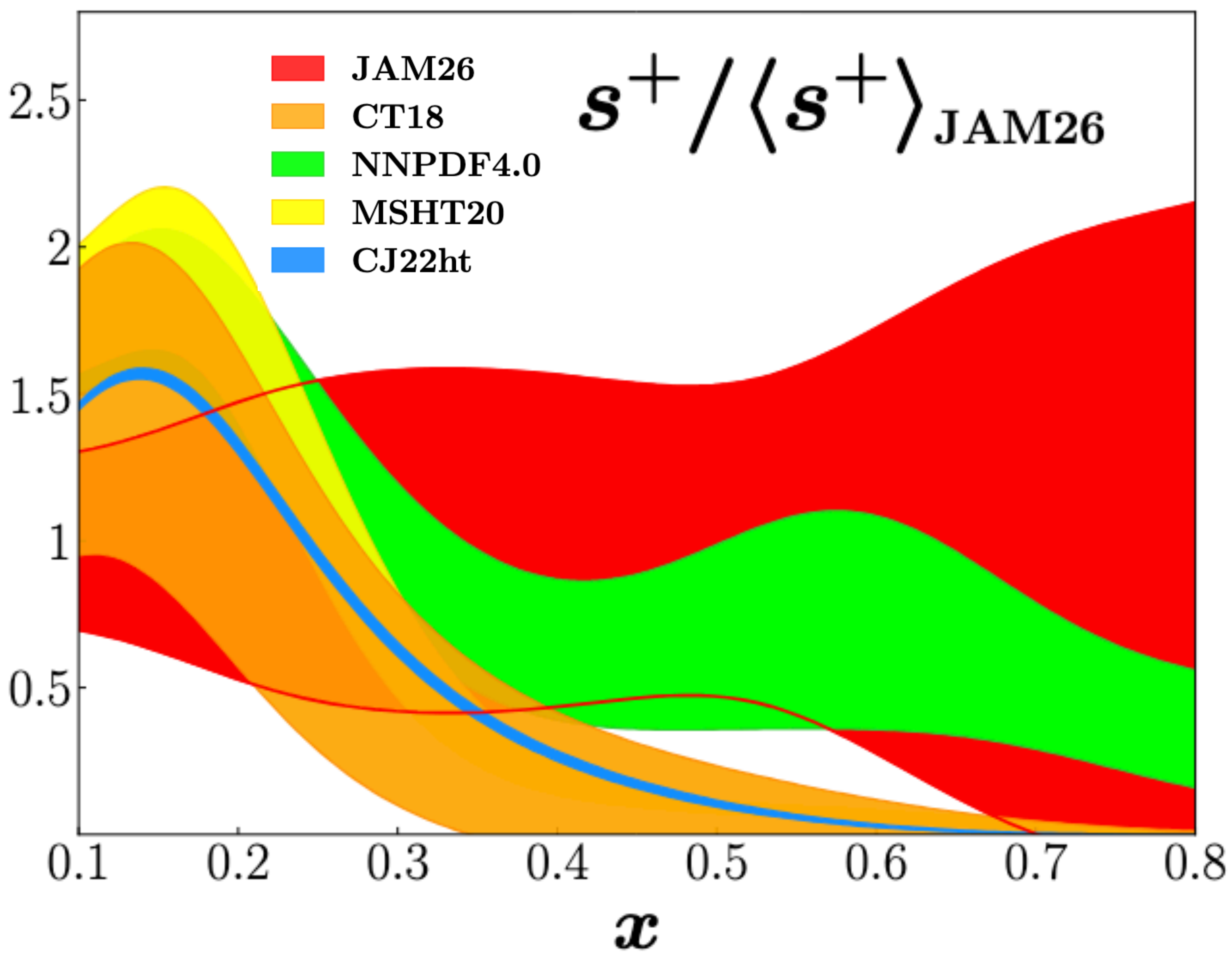


How can we improve the low-Q^2 constraints on $\sin^2\theta_\mathrm{W}$?

How strange is the proton?



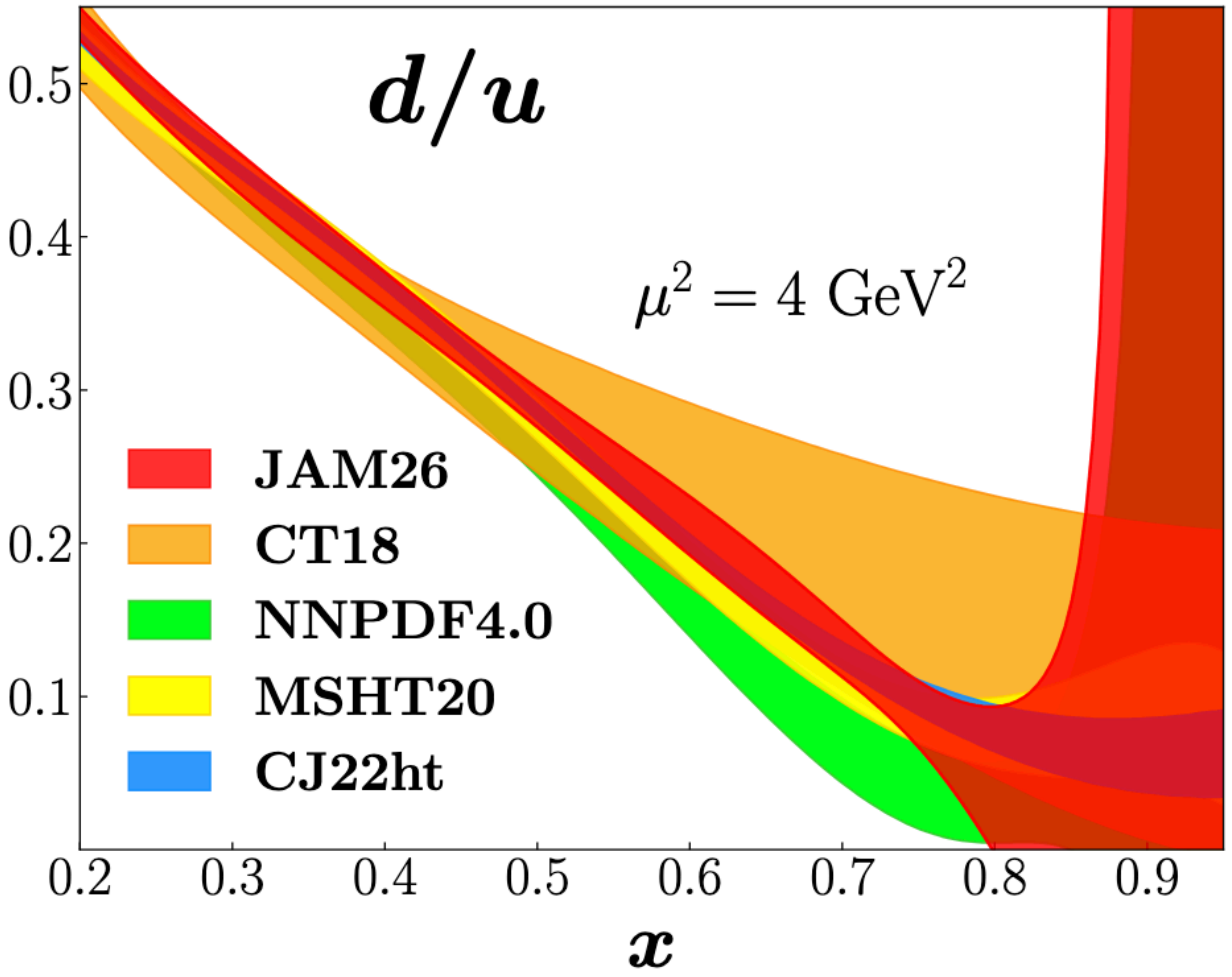
Anderson, Sato, Melnitchouk, arXiv:2501.00665v2 [hep-ph] (2025)



→ size of the strange PDFs? — s^+

→ strange sea asymmetry? — s^-

What about d/u?



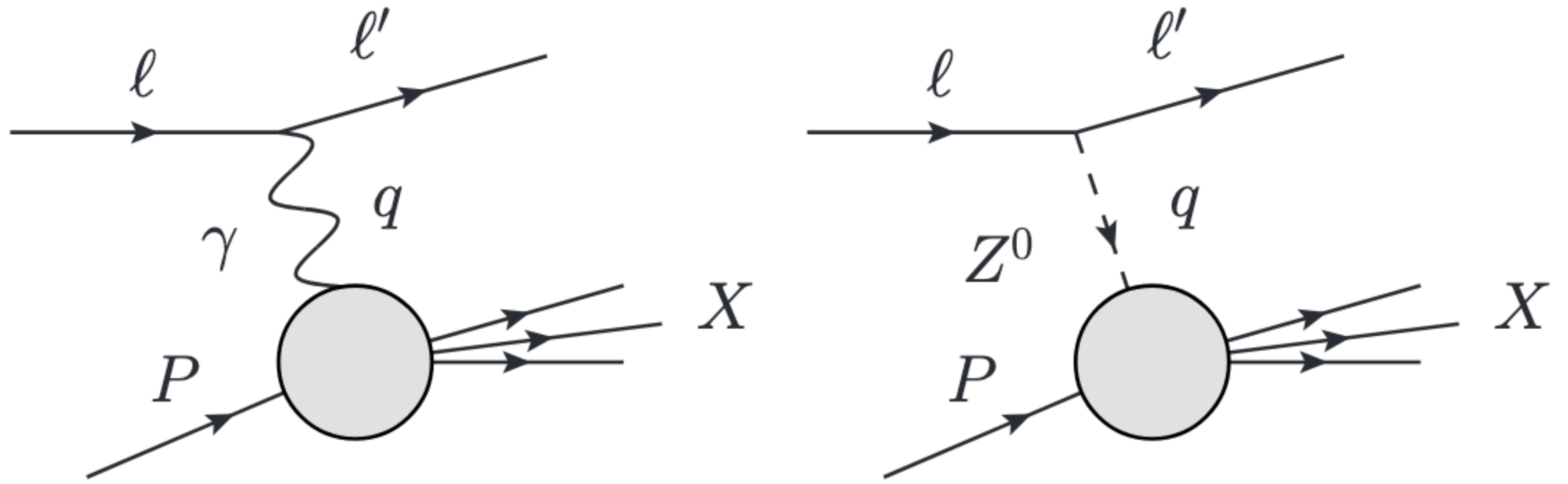
Gives NP valence/constituent structure as $x \to 1$:

$\to$ can't exclude many models with current errors!

Existing data hide signal with other effects:

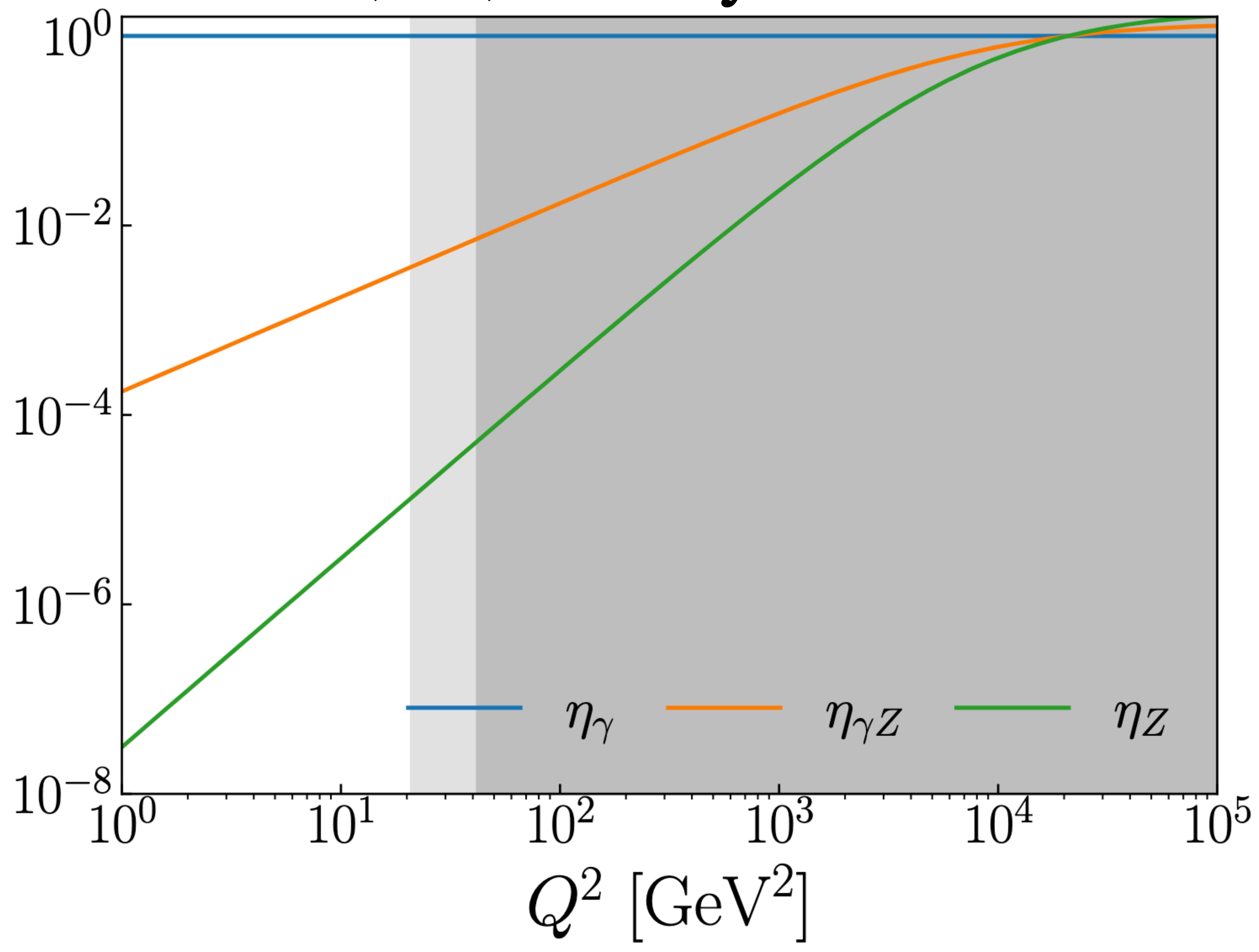
$\to$ predominately nuclear modifications!

Parity-violation in DIS



$$\frac{d\sigma_{\lambda_\ell}}{dx\,dy} = \frac{2\pi\alpha^2 y}{Q^2} \sum_{j\in\{\gamma,\gamma Z,Z\}} \eta_j C_j\, L^{\gamma}_{\mu\nu} W^{\mu\nu}_{j,U}$$

JLab11, 22, and beyond →



Parity-violating asymmetry

$$\boxed{A_{\mathrm{PV}} = \frac{\mathrm{d}\sigma_+ - \mathrm{d}\sigma_-}{\mathrm{d}\sigma_+ + \mathrm{d}\sigma_-} \approx \frac{G_F Q^2}{4\sqrt{2}\pi\alpha}\left[2g_A^e \frac{F_1^{\gamma Z}}{F_1^\gamma} Y_1 + g_V^e \frac{F_3^{\gamma Z}}{F_1^\gamma} Y_3\right]}$$

$$g_A^e = -1/2, \quad g_V^e = -1/2 + 2\sin^2\theta_{\mathrm{W}}$$

$$Y_1 = \left(\frac{1+R^{\gamma Z}}{1+R^\gamma}\right)\frac{1+(1-y)^2 - \frac{y^2}{2}\left[1+r^2-\frac{2r^2}{1+R^{\gamma Z}}\right]}{1+(1-y)^2 - \frac{y^2}{2}\left[1+r^2-\frac{2r^2}{1+R^{\gamma}}\right]}, \quad r^2 = 1 + 4M^2x^2/Q^2$$

$$Y_3 = \left(\frac{1+R^{\gamma Z}}{1+R^\gamma}\right)\frac{1-(1-y)^2}{1+(1-y)^2 - \frac{y^2}{2}\left[1+r^2-\frac{2r^2}{1+R^{\gamma}}\right]}, \quad R^i = \frac{F_2^i}{2xF_1^i}r^2 - 1$$

Parity-violating asymmetry

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Parity-violating asymmetry

$$\boxed{A_{\mathrm{PV}} \approx \frac{G_F Q^2}{4\sqrt{2}\pi\alpha}\left[2g_A^e \frac{F_1^{\gamma Z}}{F_1^{\gamma}} Y_1 + g_V^e \frac{F_3^{\gamma Z}}{F_1^{\gamma}} Y_3\right]}$$

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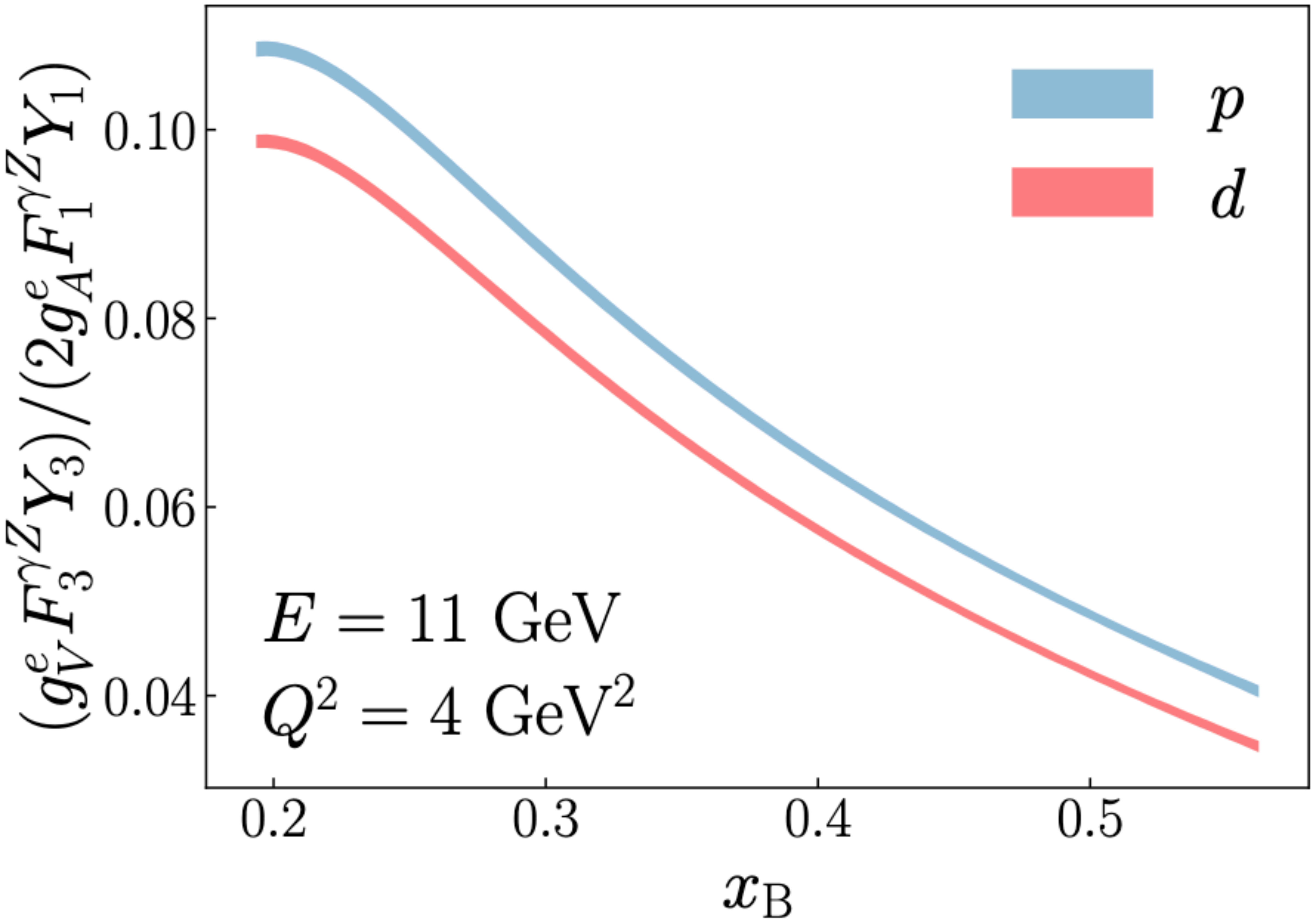
$$Y_1 \approx 1, \quad Y_3 \approx \frac{1-(1-y)^2}{1+(1-y)^2}$$

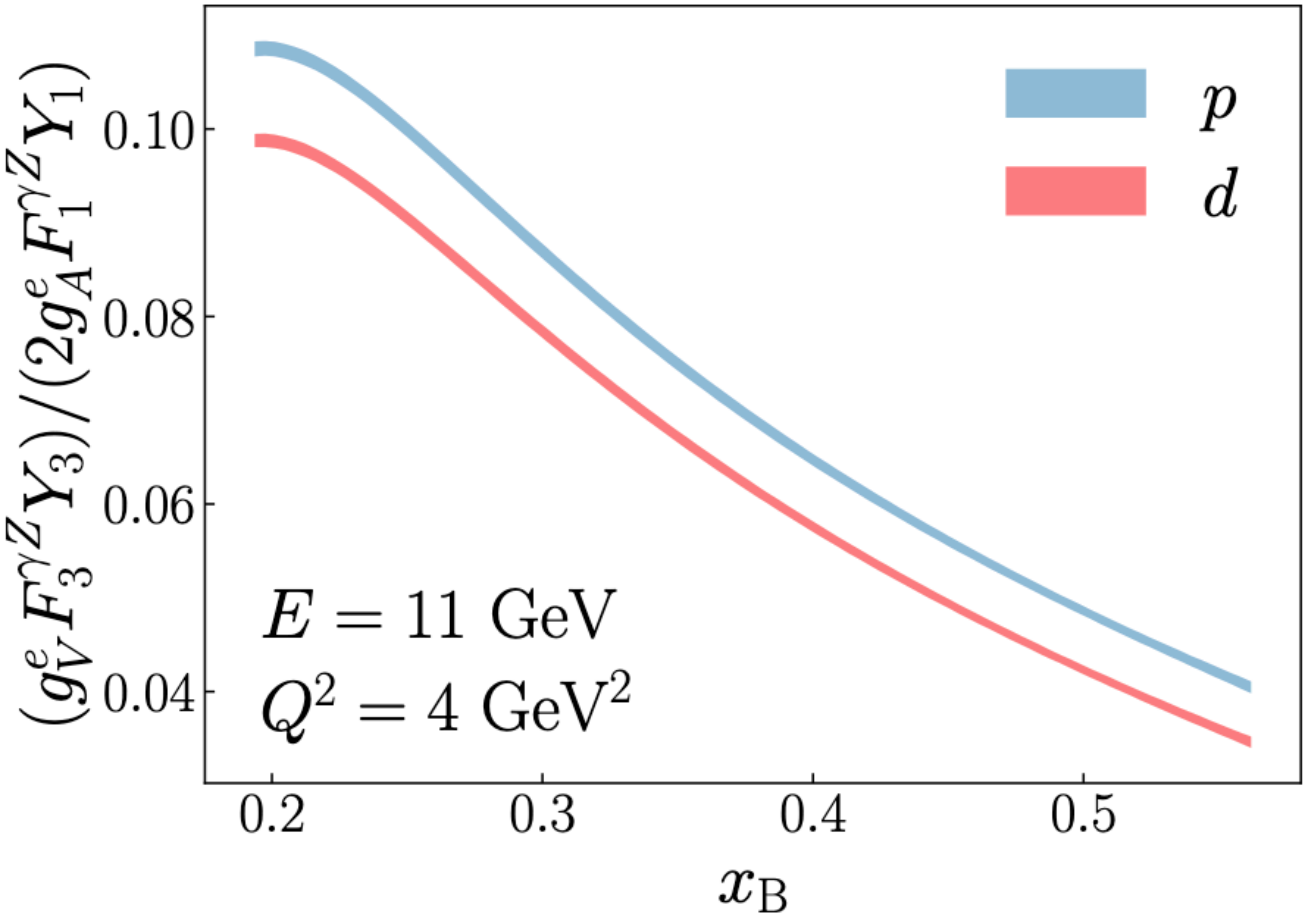


Parity-violating asymmetry

$$\boxed{A_{\mathrm{PV}} \approx -\frac{G_F Q^2}{4\sqrt{2}\pi\alpha}\frac{\sum_q 2e_q g_V^q q^+}{\sum_q e_q^2 q^+}}$$

$$g_A^e = -1/2, \quad g_V^e = -1/2 + 2\sin^2\theta_{\mathrm{W}}$$

$$Y_1 \approx 1, \quad Y_3 \approx \frac{1-(1-y)^2}{1+(1-y)^2}$$



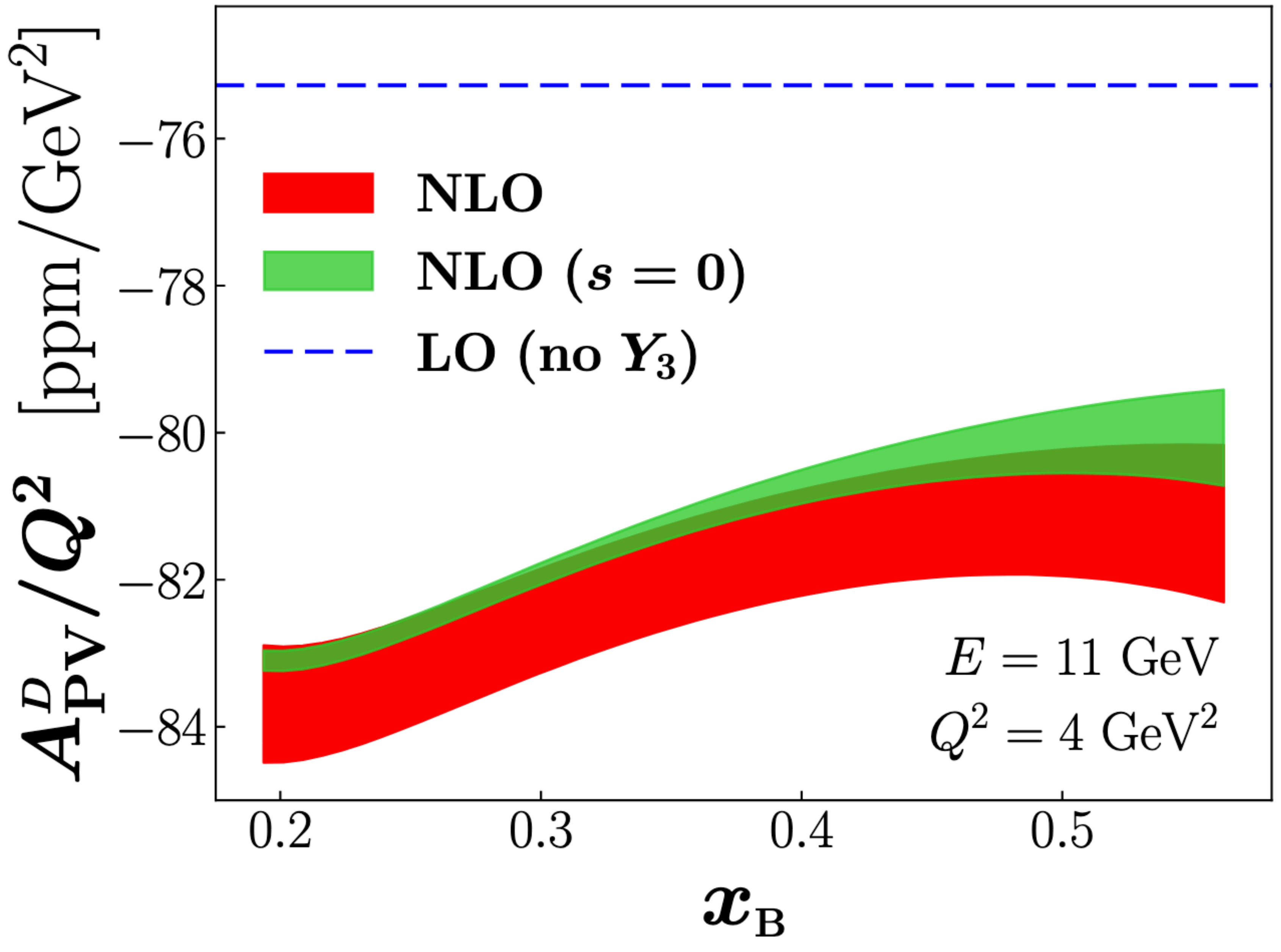
Sensitivities from deuteron and proton

$$A^D_{\rm PV} \approx -\frac{G_F Q^2}{4\sqrt{2}\pi\alpha}\left[\left(\frac{9}{5} - 4\sin^2\theta_{\rm W}\right) + \frac{2}{25}\frac{s^+}{u^+ + d^+}\right]$$

→ Dominant weak mixing signal

→ PDF/pQCD input is important

→ strangeness provides large uncertainty



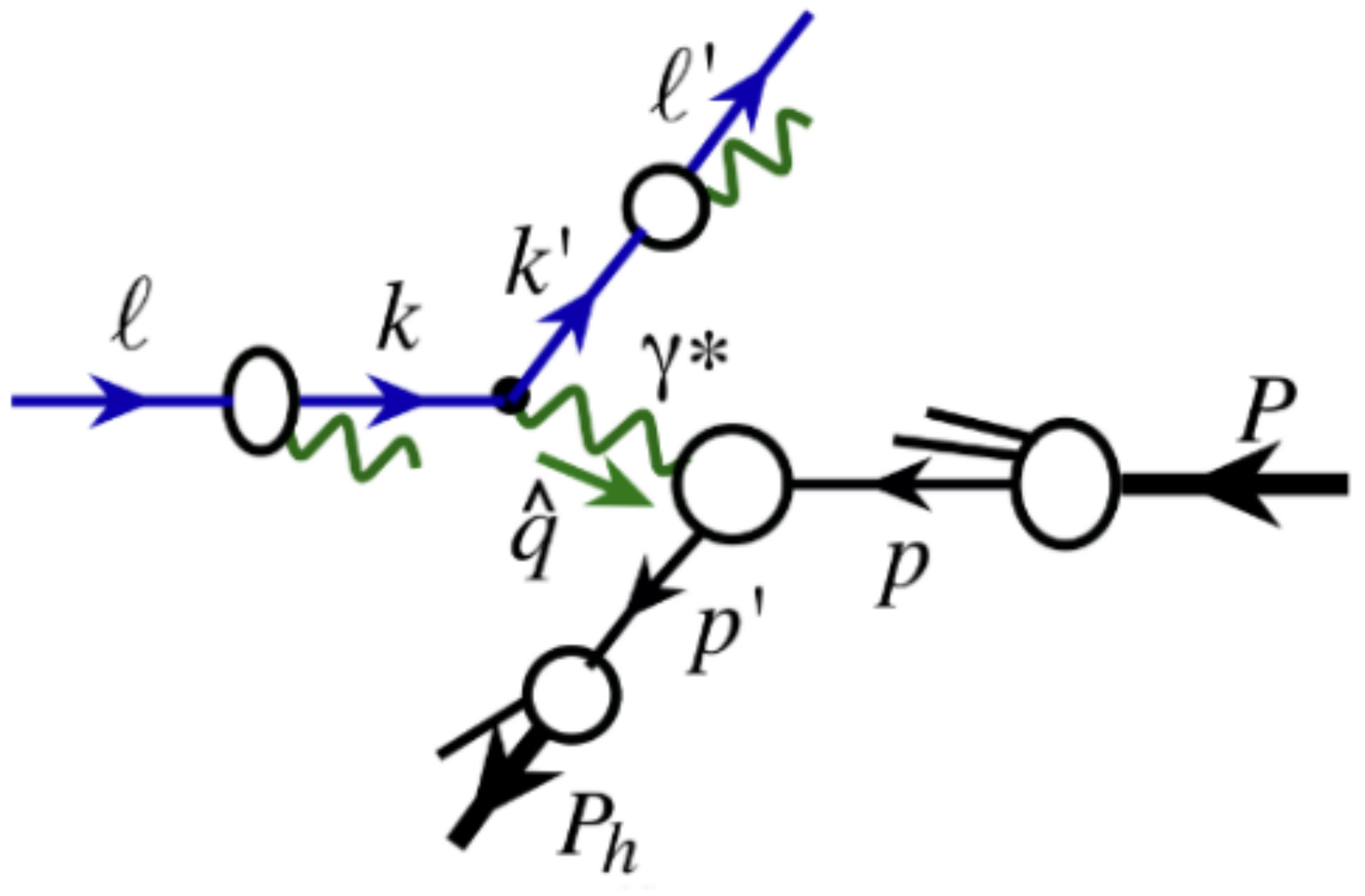
$$A^p_{\rm PV} \approx -\frac{G_F Q^2}{4\sqrt{2}\pi\alpha}(3g_V^u)\frac{1 - \frac{g_V^d}{2g_V^u}d/u}{1 + \frac{1}{4}d/u}$$

→ d/u without nuclear corrections

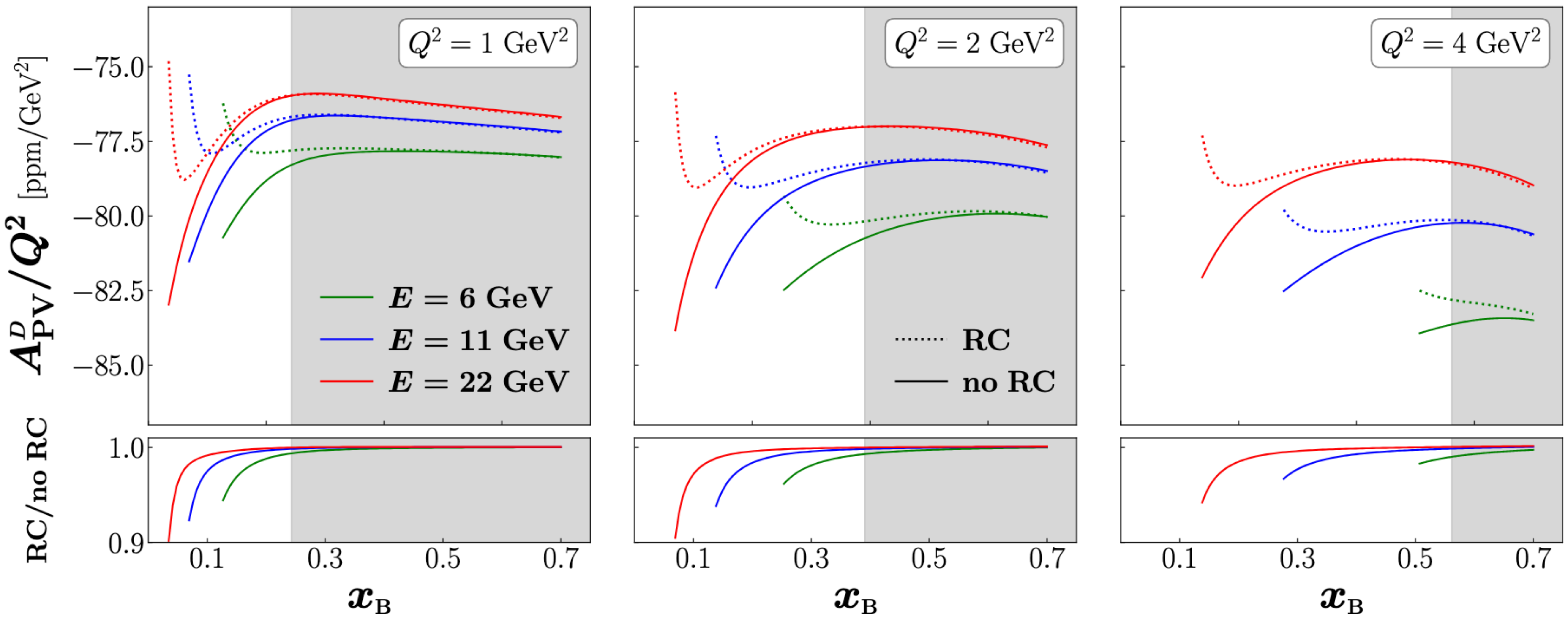
→ EW/QCD coupling multiplicative ... less correlation?

QED radiative effects

$$\frac{d\sigma}{dx\,dy} = \int_{\zeta_{\min}}^{1} \frac{d\zeta}{\zeta^2} \underbrace{D_{e/e}(\zeta,\mu^2)}_{\text{LFF}} \int_{\xi_{\min}}^{1} d\xi \underbrace{f_{e/e}(\xi,\mu^2)}_{\text{LDF}} \left[\frac{Q^2}{x}\frac{\hat{x}}{Q^2}\right] \frac{d\hat{\sigma}}{d\hat{x}\,d\hat{y}}$$



T. Liu *et al.*, JHEP 11 (2021)



Higher twist effects

Additive HT's

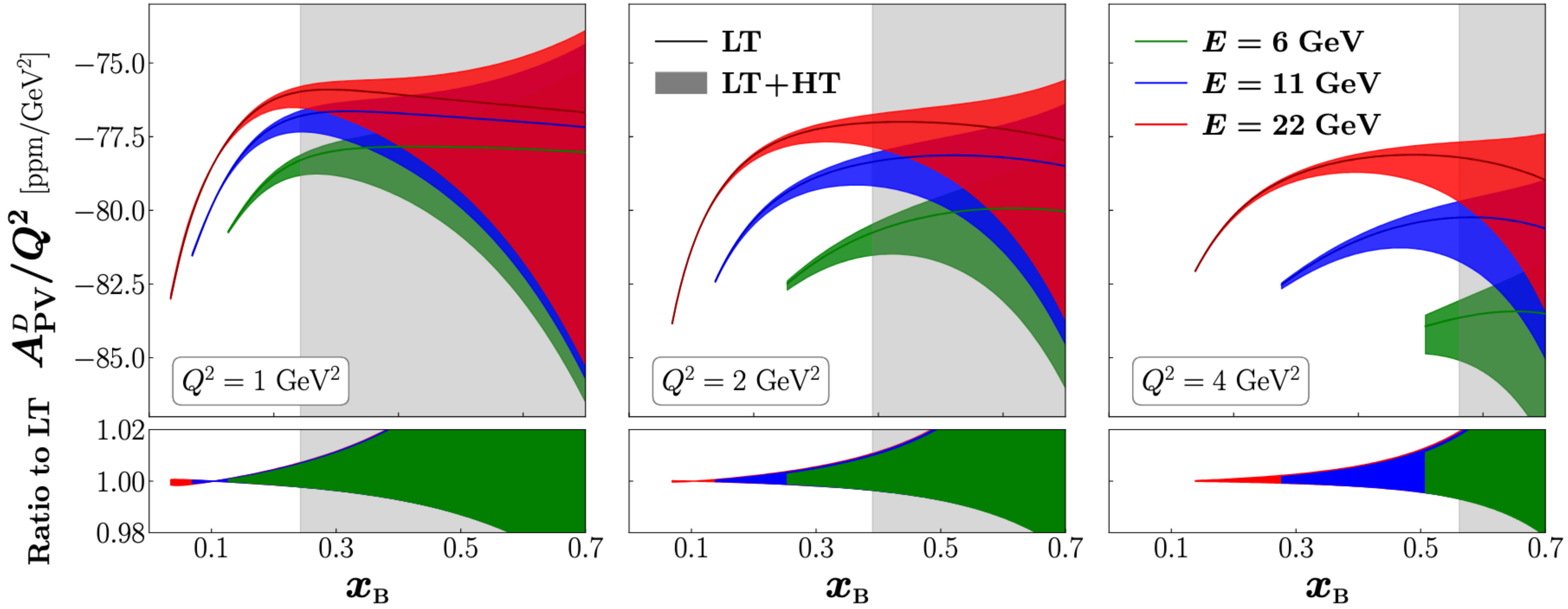
$$F_i^{\gamma} = F_{i,LT}^{\gamma} + \frac{H_i^{\gamma}(x_B)}{Q^2}$$

C. Cocuzza *et al.*, arXiv:2605.00666 [hep-ph]

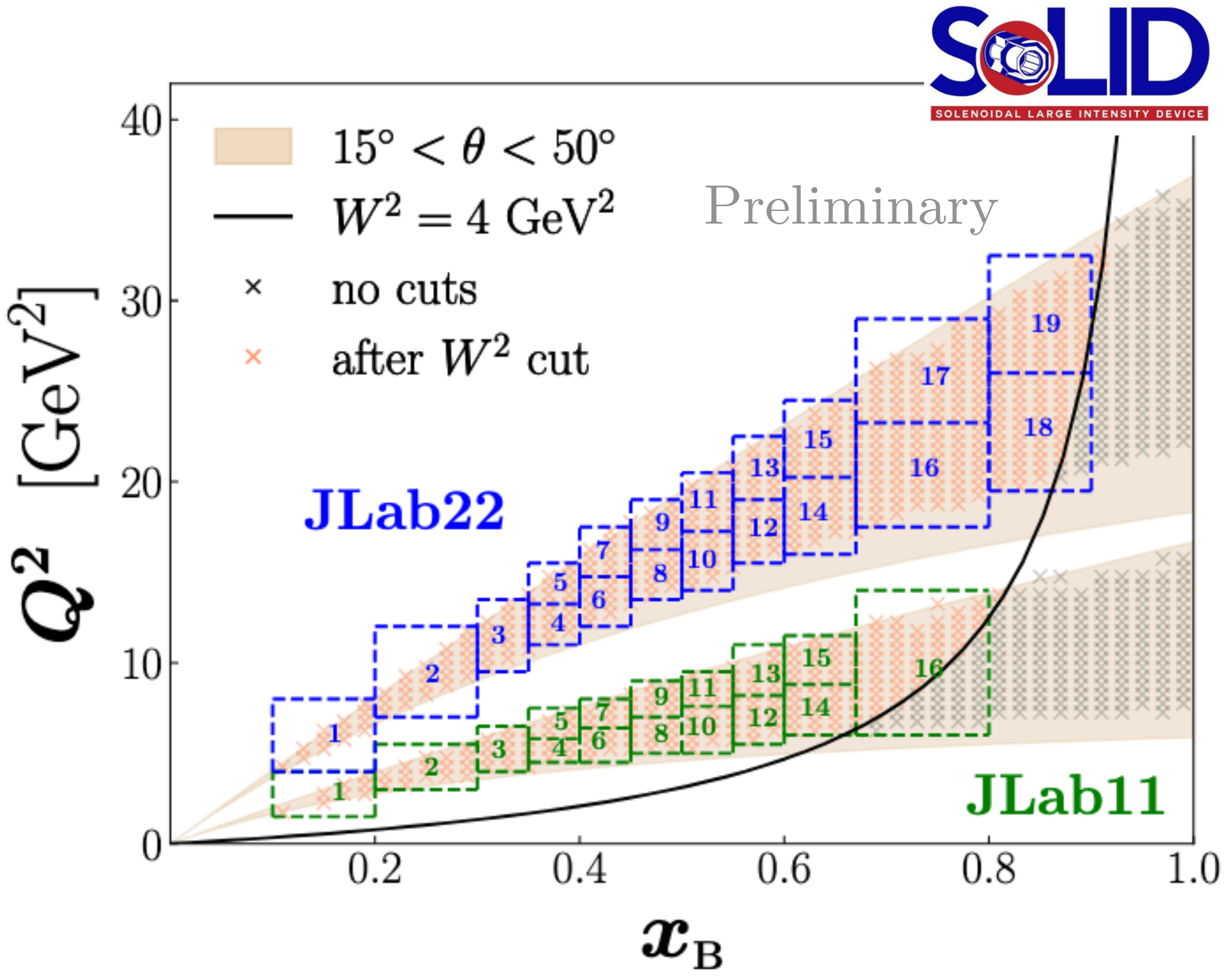
$$F_i^{\gamma Z} = F_{i,LT}^{\gamma Z} + \frac{H_i^{\gamma Z}(x_B)}{Q^2} \quad ???$$

Model

$$\rightarrow H_i^{\gamma Z} = R\, H_i^{\gamma}$$



Simulating pseudodata



Scenarios

1. stat. + exp. syst. uncertainties
2. (1) + QED effects
3. (2) + HT effects

Experimental configuration

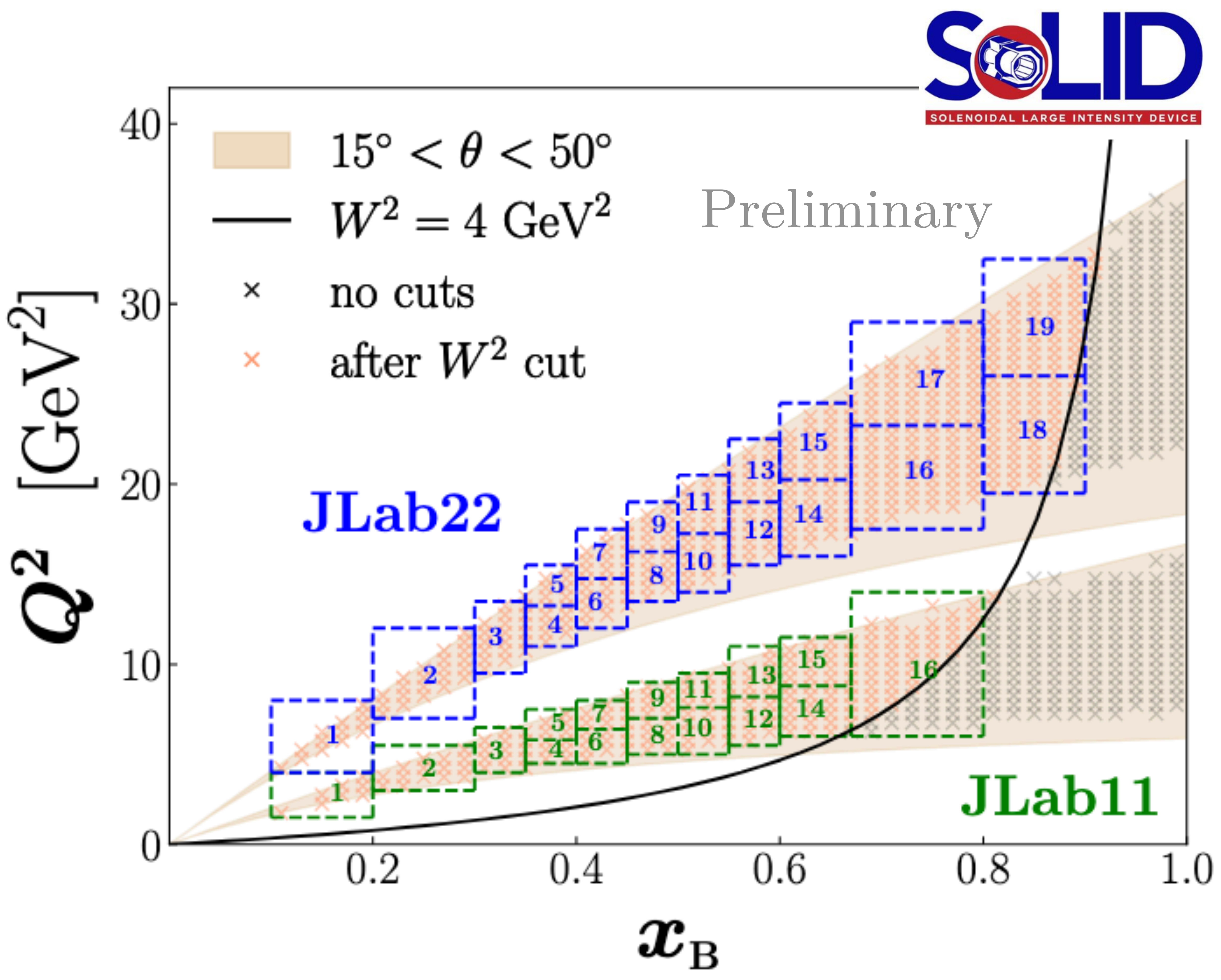
$\rightarrow P = 85\,\%$

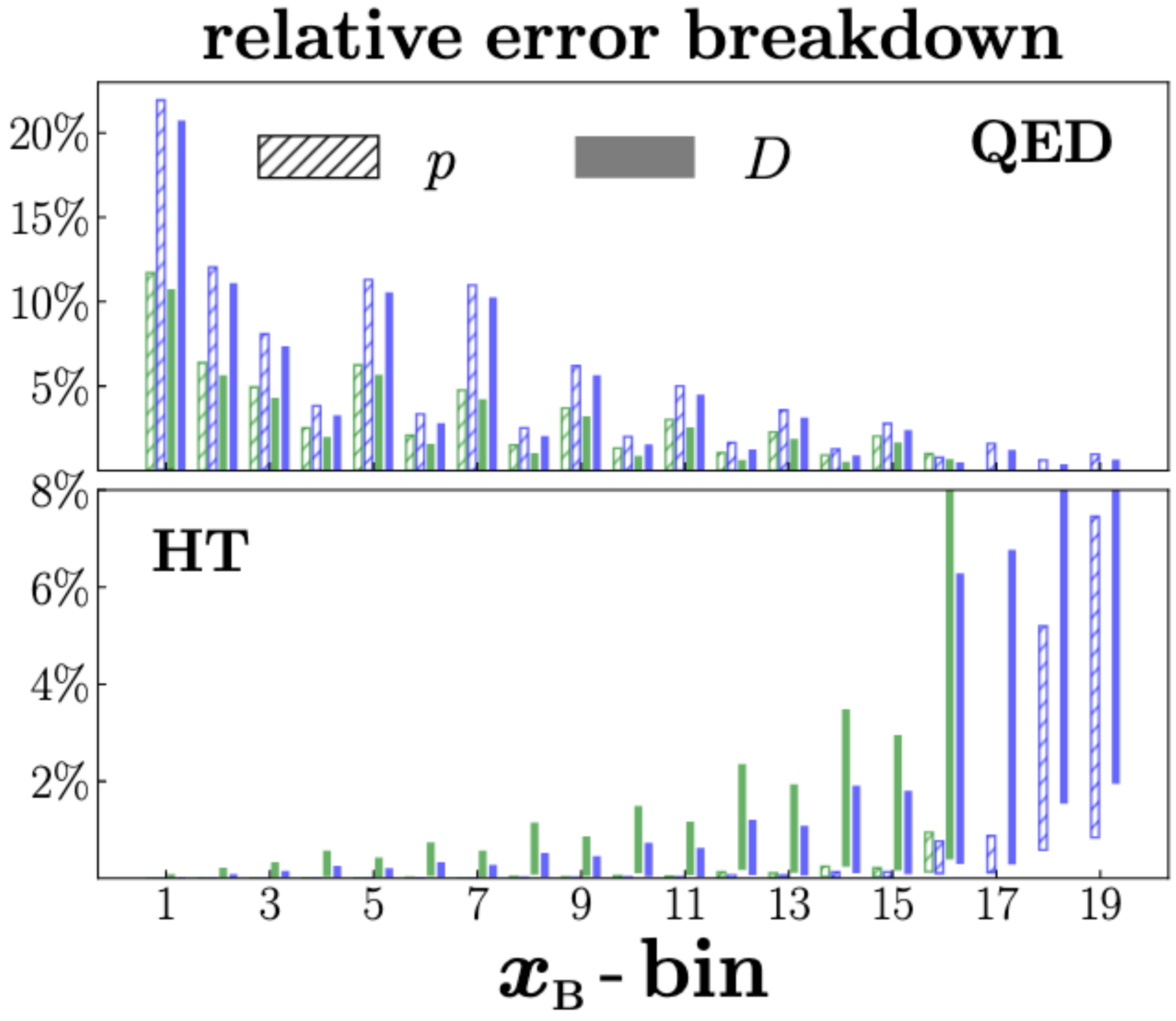
$\rightarrow \mathrm{d}\mathscr{L}/\mathrm{d}t = 4.85 \times 10^{38}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$

→ run time: 50 days/target

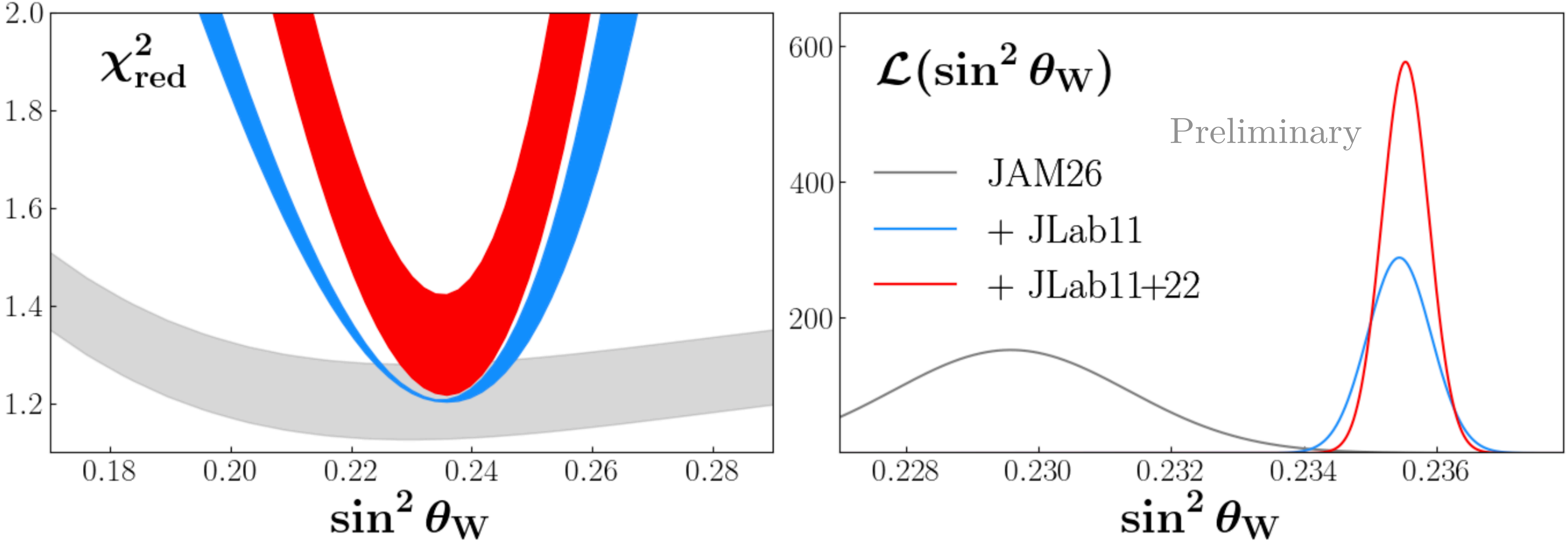
$\rightarrow \delta^{\mathrm{syst}} A_{\mathrm{PV}} = 0.5\,\%$

Simulating pseudodata





Scanning for weak mixing



JAM global analysis framework

PDF template model

$$f(x;\mathbf{a}) = \frac{N}{M} x^a (1-x)^b (1 + c\sqrt{x} + dx)$$

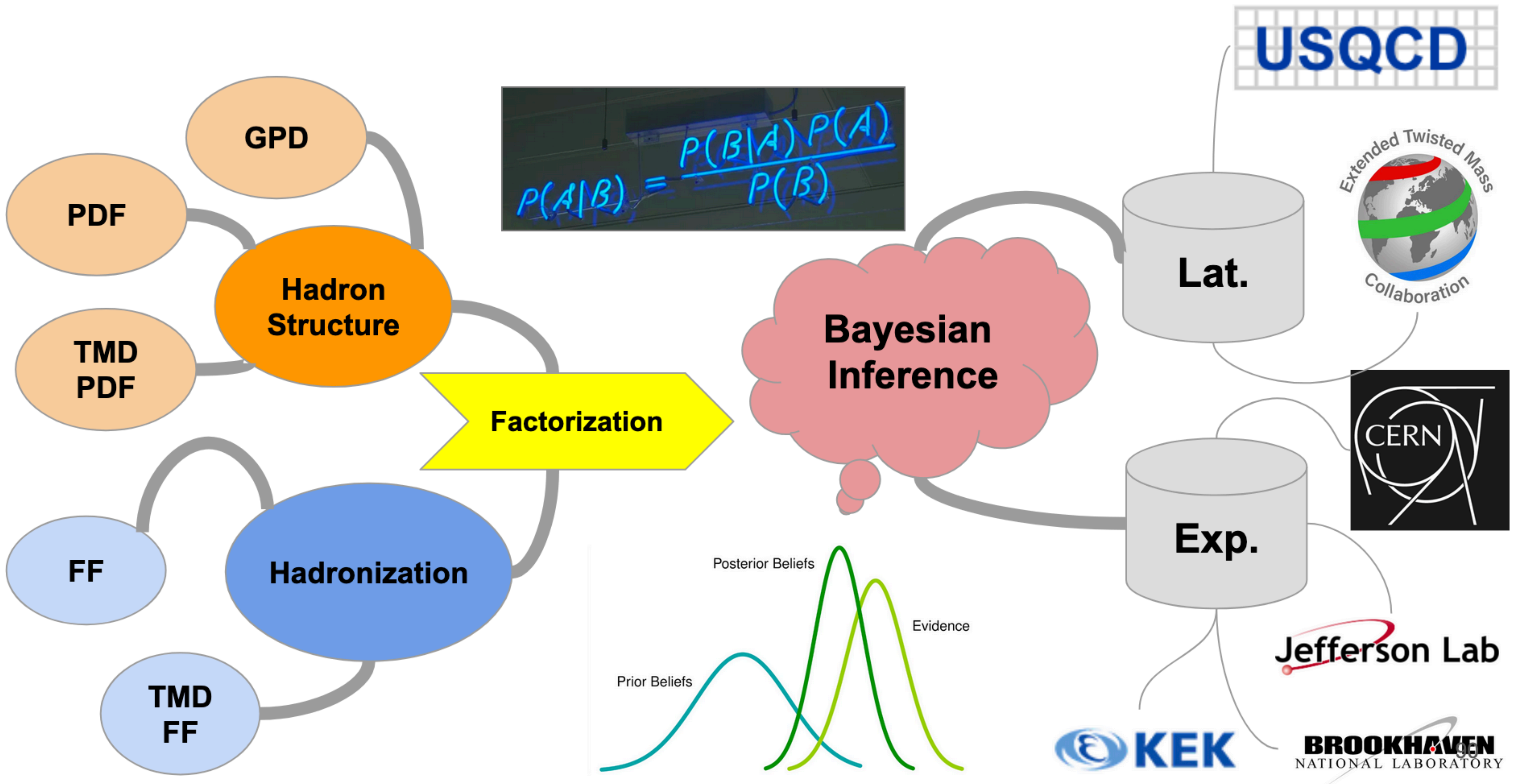
Replica PDFs

$$\mathbf{a}^{(k)} = \underset{\mathbf{a}}{\operatorname{argmin}}\, \chi^2(\mathbf{a}, \widetilde{\mathscr{D}}^{(k)})$$

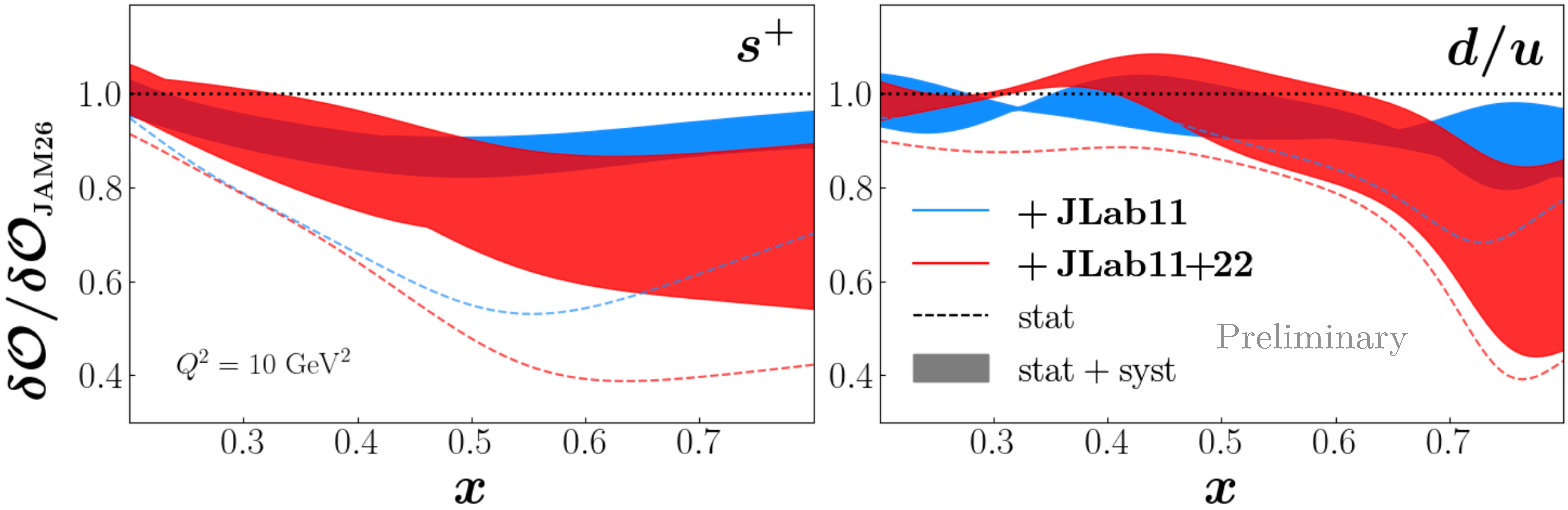
Sample statistics

$$\mathrm{E}[\mathscr{O}] = \frac{1}{n} \sum_k \mathscr{O}(\mathbf{a}^{(k)})$$

$$\mathrm{V}[\mathscr{O}] = \frac{1}{n} \sum_k \left(\mathscr{O}(\mathbf{a}^{(k)}) - \mathrm{E}[\mathscr{O}]\right)^2$$

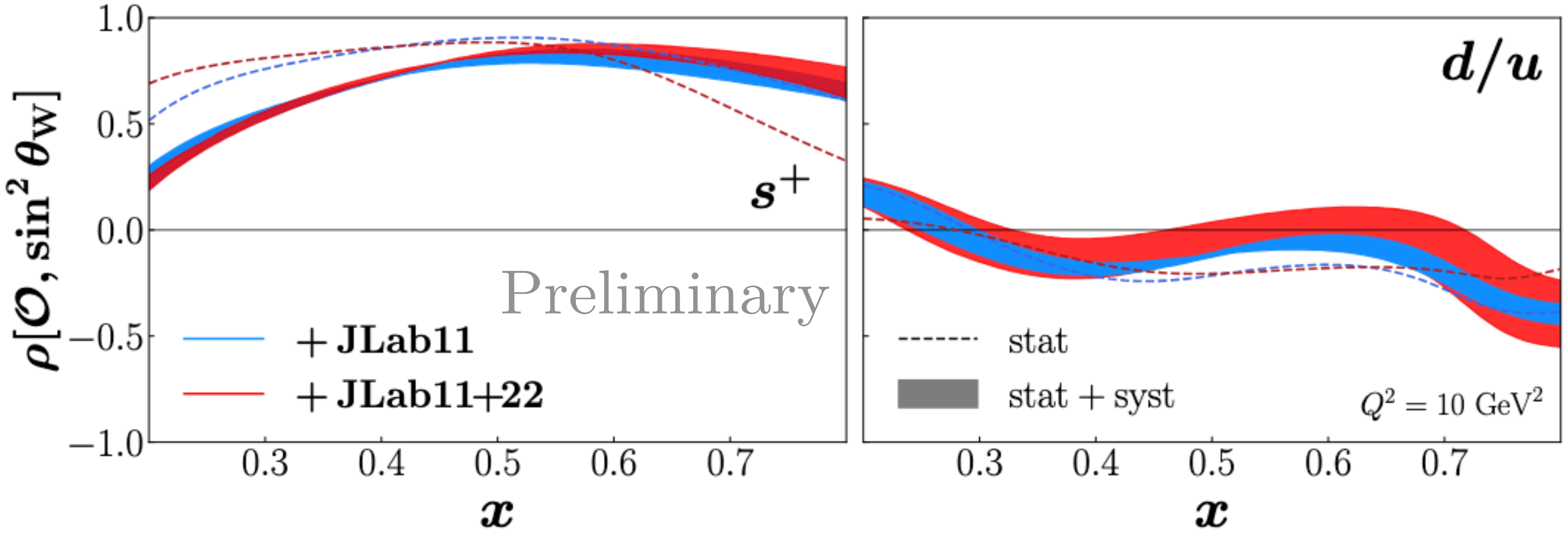


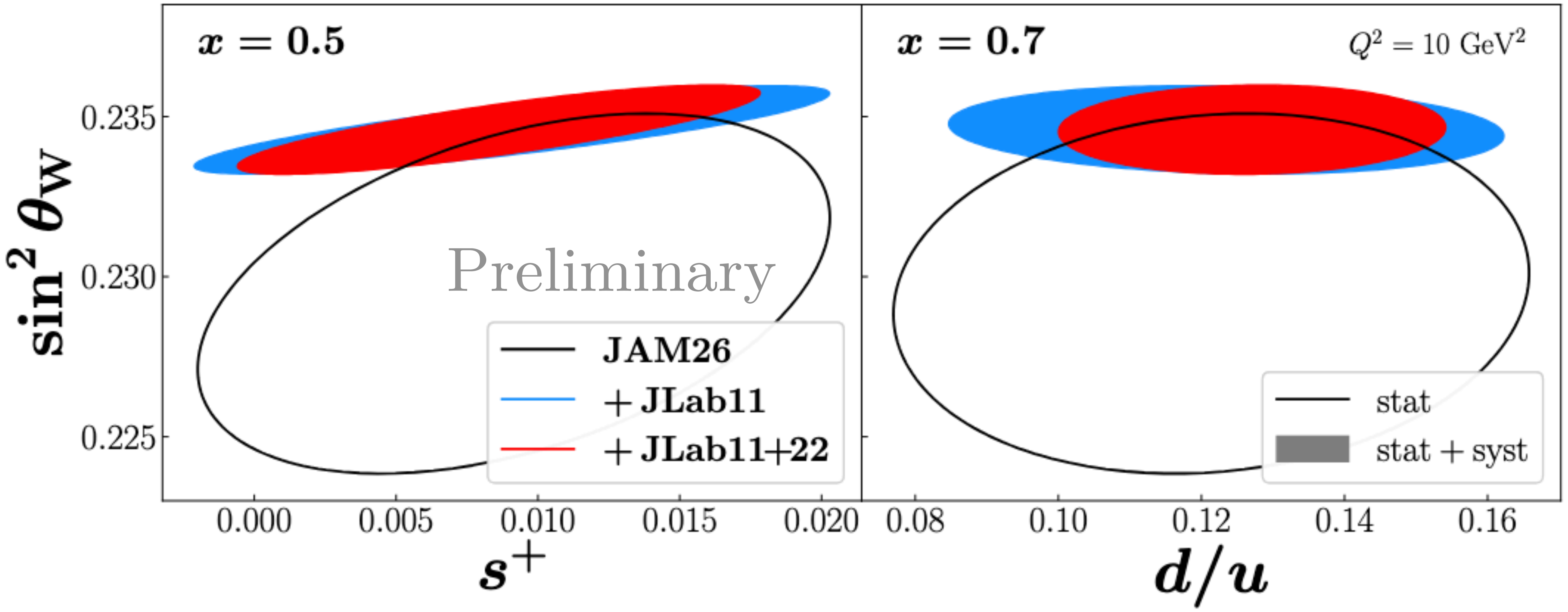
Impact on PDFs



EW & PDF correlations

Correlations can be significant!





Robust extractions of EW with QCD background requires simultaneous analyses!

What kinematics drive constraints?

Mapping data sensitivities in global QCD analysis with linear response and influence functions

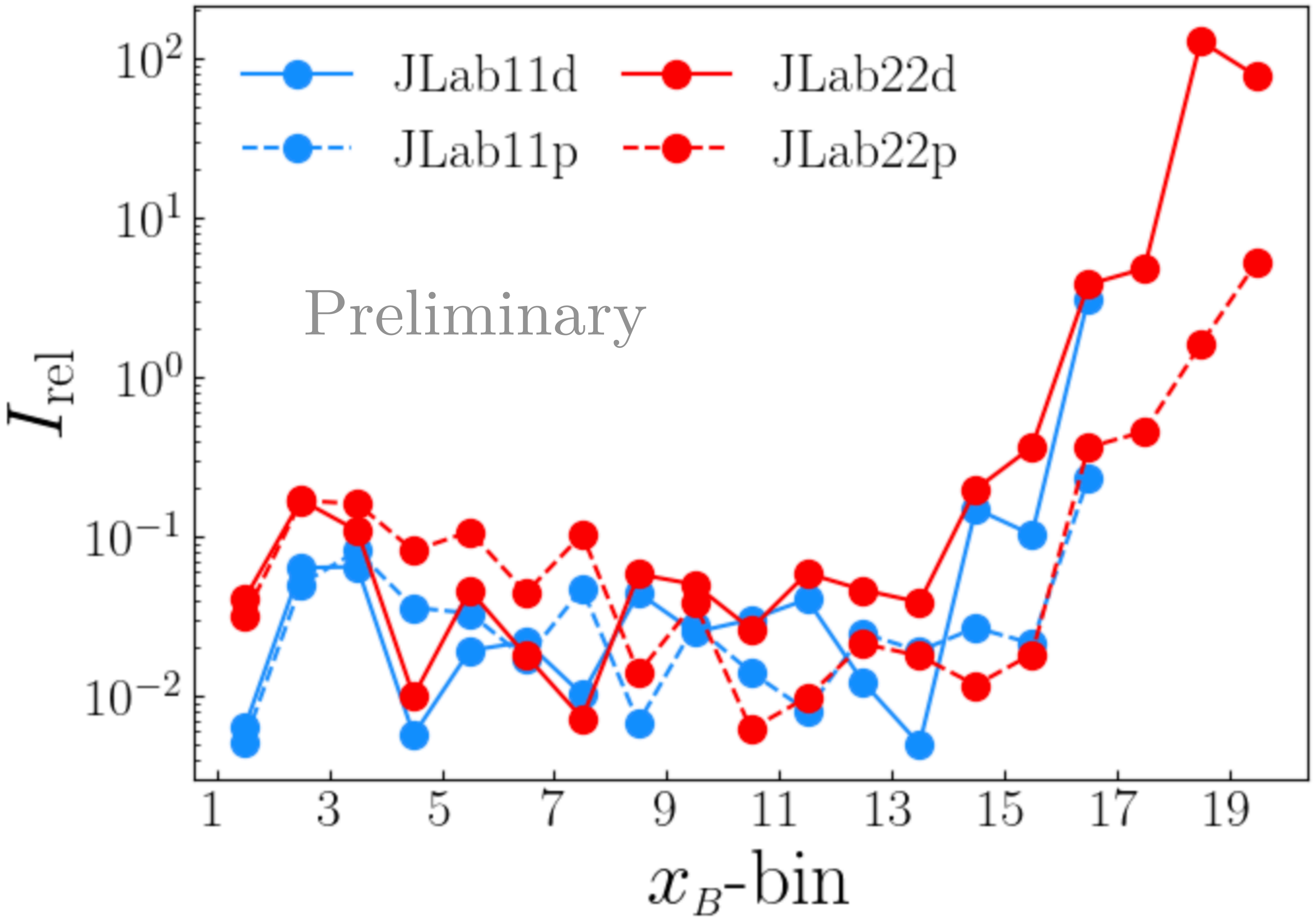
R. M. Whitehill[1,*]

$$(\chi^2)' = \chi^2 - \epsilon \chi_j^2$$

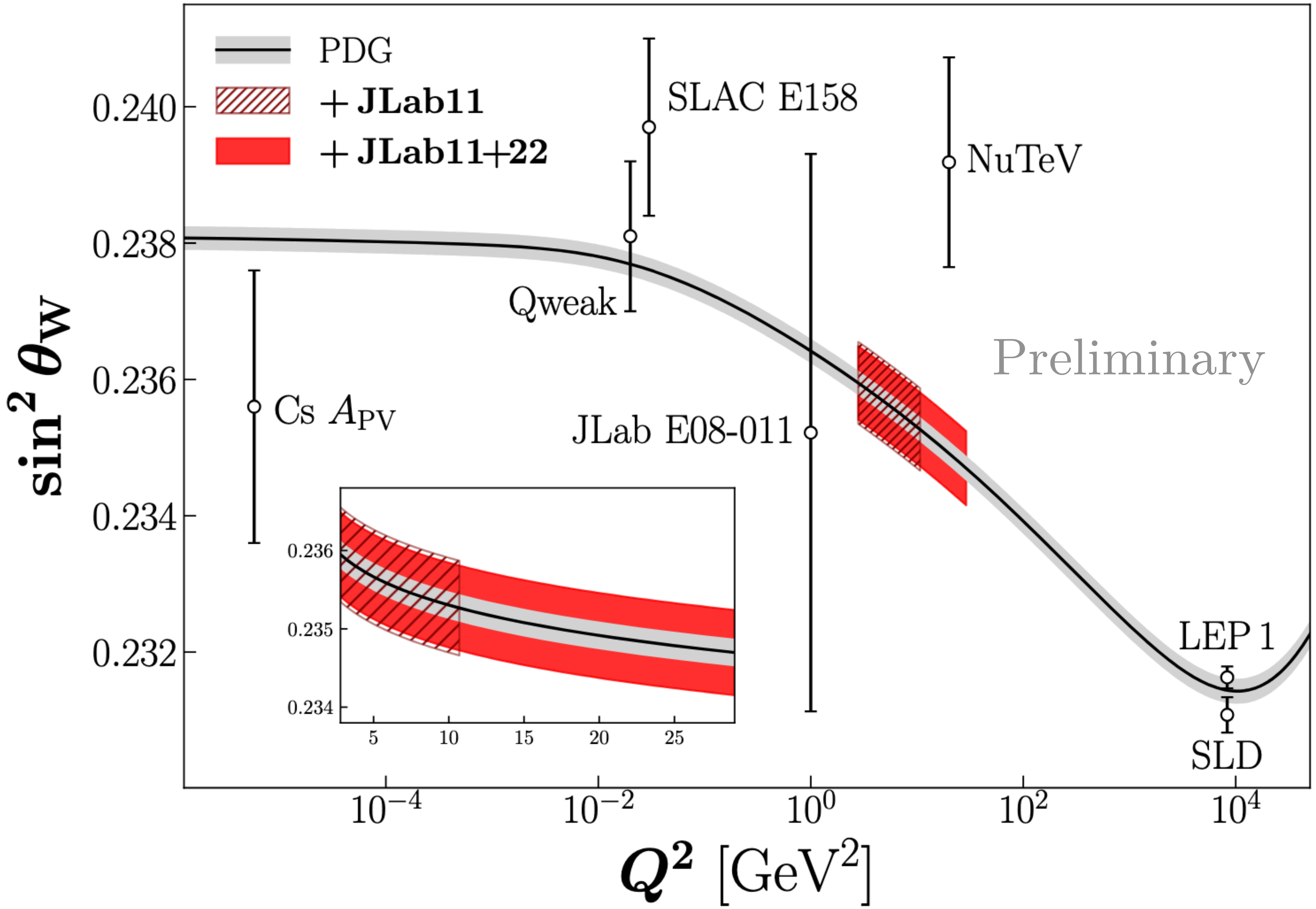
$$\Rightarrow I_{\mathrm{rel}} = \frac{1}{\delta s_{\mathrm{W}}^2} \frac{\mathrm{d}[\delta s_{\mathrm{W}}^2]}{\mathrm{d}\epsilon}$$

$$= \frac{1}{4(\delta s_{\mathrm{W}}^2)^2} \mathrm{Cov}\left[(s_{\mathrm{W}}^2 - \langle s_{\mathrm{W}}^2 \rangle)^2, \chi_j^2\right]$$

SLAC + JLab6 A_{PV} (legacy): $I_{\mathrm{rel}} \approx -3\,\%$



Simultaneous EW & PDF impacts



A_{PV} from JLab can provide most precise & unbiased test of SM physics at low-Q^2 through global analysis

Conclusions & Outlook

A_{PV} is a unique/ clean observable for future global analyses:

1) tests of BSM physics through the determination of the weak-mixing angle

2) constraint of nucleon strangeness

- Robust extraction of EW physics/hadron structure requires simultaneous analysis
- Need to better understand/mitigate QED, HT effects to reduce limiting uncertainties
- Moderate-x_B & Q^2 data is best … $22\ \mathrm{GeV}$ electrons?
- Sea asymmetry from complimentary positron data?
- Polarized strangeness at EIC?